Elementary & Intermediate Algebra

Solving linear equations

1 Solve for x: $4x - 9 = 23$.

2 Solve for x: $\frac{3x - 2}{5} = 8$.

Solving and graphing inequalities

3 Solve $5x + 3 < 28$, giving your answer as an inequality.

4 Solve $-4x + 7 \leq -9$, being careful with the direction of the inequality.

Systems of linear equations

5 Solve the system: $y = 3x - 4$ and $2x + y = 11$.

6 Solve the system: $3x + 4y = 18$ and $x - 2y = -4$, using elimination.

Factoring and solving quadratic equations

7 Solve $x^2 - x - 12 = 0$ by factoring.

8 Solve $2x^2 - 5x - 3 = 0$ by factoring.

Simplifying rational and radical expressions

9 Simplify $\frac{x^2 - 9}{x + 3}$.

10 Simplify $\sqrt{48}$.

Working with exponents

11 Simplify $\frac{x^7}{x^3}$.

12 Simplify $(3x^2y)^3$.

Absolute value equations

13 Solve $|3x+2|=11$.

Word problems leading to linear or quadratic equations

14 The length of a rectangle is 3 more than twice its width, and its area is 65. Find the width and length.

15 A number increased by 8 is 3 times the original number decreased by 4. Find the number.

Combining algebraic techniques in a multi-step problem

16 This question has two parts. Two consecutive positive integers have a product of 132. (a) Set up and solve a quadratic equation to find the two integers. (b) Verify your answer by checking the product.

Solving quadratics using the quadratic formula

17 Solve $x^2+3x-2=0$ using the quadratic formula, giving exact answers.

18 Use the discriminant to determine the number of real solutions of $4x^2-12x+9=0$, without solving.

Factoring by grouping and special patterns

19 Factor $x^3+2x^2-9x-18$ by grouping.

20 Factor $4x^2-25$ completely.

Rational equations and extraneous solutions

21 Solve $\frac{x}{x-2}=\frac{3}{x-2}+1$, checking for extraneous solutions.

Multi-step algebraic word problems

22 This question has two parts. A boat travels 60 miles downstream in 3 hours and returns upstream in 5 hours. (a) Let b be the boat's speed in still water and c be the current's speed; write two equations based on the trip times. (b) Solve the system to find b and c.